

Fast and Accurate Online Coupled Matrix-Tensor Factorization via Frequency Regularization

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Abstract

How can we efficiently and accurately factorize multi-source data in dynamic and real-time environments? Coupled matrix-tensor factorization (CMTF) is a powerful tool for such tasks, but existing methods often struggle with scalability, particularly when dealing with continuously streaming data. Traditional CMTF approaches, while effective at capturing complex relationships, suffer from computational inefficiencies and the need for retraining as new data arrive. Moreover, many techniques fail to properly incorporate the inherent temporal characteristics of the data, which could significantly enhance both accuracy and convergence speed.

In this paper, we propose FOCAL (Frequency-regularized Online Coupled Approximation for Low-rank factorization), an efficient CP decomposition method designed to enhance online coupled matrix-tensor factorization. By effectively distinguishing between old and new data, FOCAL optimizes computational efficiency, reducing redundant computations and eliminating the need for full retraining in streaming settings. Furthermore, FOCAL integrates frequency regularization into an online CMTF framework, which mitigates overfitting and improves accuracy. Through extensive experiments, we demonstrate that FOCAL outperforms existing state-of-the-art methods in terms of both speed and accuracy. We also present results on anomaly detection using real-world data, showcasing FOCAL's effectiveness in identifying irregular patterns.

CCS Concepts

• **Information systems** → **Data stream mining**; • **Computing methodologies** → **Factorization methods**; *Anomaly detection*; • **Mathematics of computing** → *Dimensionality reduction*; *Time series analysis*.

Keywords

Streaming tensor, Coupled matrix-tensor factorization, Frequency regularization, Time series analysis, Anomaly detection

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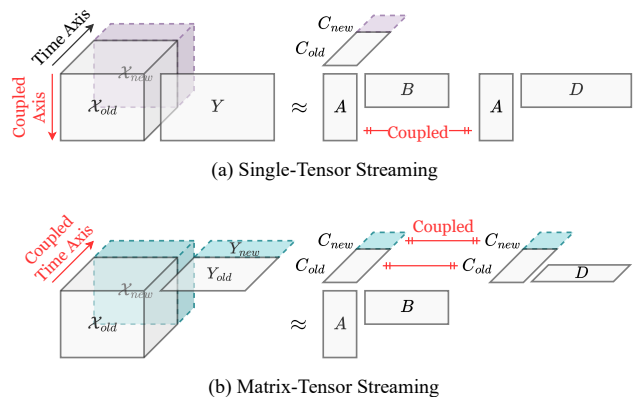


Figure 1: Two distinct streaming scenarios in CMTF. (a) In single-tensor streaming, only the main tensor evolves over time as new slices of data arrive, while the matrix remains constant. (b) In matrix-tensor streaming, both the tensor and the matrix sharing a coupled temporal dimension are concurrently updated with incoming data.

1 Introduction

The surge of multi-modal data in modern AI applications—including data mining [16, 21, 34, 35, 41, 42], recommender systems [20, 30–32, 51], graph learning [24, 26, 27], and signal processing [13, 22, 23, 25, 40]—calls for analytic methods that integrate heterogeneous sources in real time. Coupled matrix-tensor factorization (CMTF) meets this need in the *static* regime by jointly decomposing a tensor and an aligned matrix, uncovering low-rank factors that link their shared entities. Such static CMTF models already power fraud detection in finance [49], traffic optimization in smart-city platforms [6], social-media trend analysis [5], and patient monitoring [3]. Continuous data arrival in these domains makes batch decompositions computationally prohibitive and inadequate for tracking rapid temporal dynamics, motivating the necessity of an *online* CMTF formulation for real-time adaptability and accuracy. In this context, we consider two distinct streaming scenarios:

- **Setting 1: Single-Tensor Streaming.** In this setting, only the primary tensor is dynamic, with new slices of data arriving over time, while the auxiliary matrix remains static and provides a fixed reference for the coupled decomposition (see Figure 1(a)). This scenario is common in applications where the static matrix represents a stable baseline or metadata, while the streaming tensor evolves with real-time updates. The main challenge here is to efficiently update the factorization, leveraging previously computed factors to minimize redundant calculations.

• **Setting 2: Matrix-Tensor Streaming.** In this more complex scenario, both the tensor and matrix sharing a coupled dimension are simultaneously updated with streaming data, requiring consistent and synchronized factorization updates (see Figure 1(b)). For example, in sensor networks, multiple datasets recording related environmental conditions must be updated simultaneously as new measurements arrive. The challenge here is to maintain the consistency of the coupled factors across the matrix-tensor while efficiently integrating the incoming data.

Handling streaming data in coupled matrix-tensor factorization is challenging due to frequent scalability bottlenecks. Many existing methods require full retraining upon data arrival, incurring substantial computational overhead. Moreover, a critical limitation of these methods is their failure to incorporate the inherent temporal characteristics of the data. Without utilizing temporal patterns, existing techniques miss opportunities to enhance both accuracy and convergence speed, which are vital for real-time tasks.

In this paper, we propose FOCAL (Frequency-regularized Online Coupled Approximation for Low-rank factorization), an efficient and scalable CANDECOMP/PARAFAC (CP) decomposition method for online coupled matrix-tensor factorization. FOCAL adaptively handles streaming data while maintaining high decomposition quality and computational efficiency. Specifically, FOCAL offers three key features that differentiate it from current methods in CMTF:

- (1) **Adaptability.** Through an adaptive forgetting factor mechanism, FOCAL continuously adjusts its temporal focus. This allows the model to emphasize recent data without discarding informative historical trends.
- (2) **Accuracy.** FOCAL leverages a frequency-domain regularization tailored to the temporal factors; by exploiting the data’s spectral structure, it suppresses noise and mitigates overfitting, thereby improving reconstruction accuracy and stability.
- (3) **Efficiency.** FOCAL distinguishes between historical and incoming data to selectively reuse precomputed factors, minimizing redundant computation and enabling efficient streaming updates without sacrificing accuracy.

Extensive experiments on real-world datasets demonstrate the effectiveness of FOCAL. It consistently outperforms state-of-the-art baselines in both accuracy and computational efficiency. Additionally, we showcase its utility in anomaly detection tasks, where it successfully identifies irregular patterns in real-world datasets. These results highlight the robustness and versatility of FOCAL as a practical solution for real-time, large-scale data analysis.

We summarize our contributions as follows:

- **Method.** We propose FOCAL, an efficient framework for online coupled matrix-tensor factorization that incorporates adaptive forgetting and frequency regularization mechanisms.
- **Analysis.** We derive the update rules for all factor matrices, as well as the time and space complexity of updating the factors.
- **Performance.** We experimentally show that FOCAL outperforms existing approaches in terms of accuracy, convergence speed, and computational efficiency.

Table 1 summarizes the key symbols appearing in the paper. The source code and datasets used in our paper are provided at <https://github.com/snudatalab/FOCAL>.

Table 1: Symbol description.

Symbol	Description
$\mathcal{X}_{old} \in \mathbb{R}^{I \times J \times K_{old}}$	pre-existing old tensor
$\mathcal{X}_{new} \in \mathbb{R}^{I \times J \times K_{new}}$	newly arrived tensor
$Y_{old} \in \mathbb{R}^{K_{old} \times L}$	pre-existing old matrix
$Y_{new} \in \mathbb{R}^{K_{new} \times L}$	newly arrived matrix
$A \in \mathbb{R}^{I \times R}$	non-temporal factor matrix of a tensor
$B \in \mathbb{R}^{J \times R}$	non-temporal factor matrix of a tensor
$C_{old} \in \mathbb{R}^{K_{old} \times R}$	temporal factor matrix of old data
$C_{new} \in \mathbb{R}^{K_{new} \times R}$	temporal factor matrix of new data
$D \in \mathbb{R}^{L \times R}$	non-temporal factor matrix of a matrix
$\{P_n\}_{n=1}^3, \{Q_n\}_{n=1}^3$	auxiliary matrices for factor updates
$R \in \mathbb{N}$	tensor/matrix rank
$\lambda \in [0, 1]$	forgetting factor for a tensor
$\mu \in [0, 1]$	forgetting factor for a matrix
$\eta \geq 0$	learning rate of a forgetting factor
$\Phi \in \mathbb{R}^{K_{new} \times R}$	frequency filter
$\phi \in \mathbb{R}^{K_{new}}$	one-dimensional frequency mask
$\llbracket \cdot, \cdot, \cdot \rrbracket$	CP model (sum of outer products)
$\ \cdot\ _F$	Frobenius norm
$T_{(n)}$	mode- n unfolding of a tensor $\mathcal{T}$
$\Re(Z)$	real part of a complex matrix Z
$A^\dagger$	Moore–Penrose pseudoinverse of a matrix A
$*$	Hadamard product
$\odot$	Khatri-Rao product

2 Related Works

We review previous methods for streaming tensor decomposition and coupled tensor factorization.

2.1 Streaming Tensor Decomposition

Online tensor decomposition has been widely studied, with many algorithms incrementally updating factor matrices as data arrive [12, 14, 15, 50]. Streaming CP models [19, 36, 45, 46, 48] in particular have matured rapidly: OnlineCP [54] addresses dense data, OnlineSCP [53] extends the approach to sparse streams, and Sambaten [11] leverages subsampling for scalable parallel processing. Robust variants such as BRST [52] and OR-MSTC [37] further separate outliers from low-rank structure to handle missing entries.

Nevertheless, two critical challenges remain largely unresolved in the current literature. First, most streaming algorithms neglect the fact that the statistical properties of real-world tensors can evolve rapidly over time, leading to large reconstruction errors whenever an abrupt regime shift occurs. Several works [1, 38, 47] mitigate this issue by introducing an a priori forgetting factor that exponentially discounts older slices. However, because this factor is chosen *offline* and kept *fixed*, the algorithm cannot react to unpredictable accelerations or decelerations in the data’s drift, leaving it vulnerable to both sudden changes and periods of quasi-stationarity.

Second, existing methods [4, 28, 29] that explicitly exploit temporal structure typically assume a *single, well-defined pattern* (e.g., strict periodicity or local smoothness) and tailor their regularizers accordingly. While effective within the narrow domains for which they were designed, these approaches struggle when confronted with heterogeneous, overlapping temporal behaviors that commonly coexist in large-scale, real-world streams.

In contrast, FOCAL tackles both challenges simultaneously. It learns the forgetting factor on the fly by optimizing a data-driven,

adaptive objective, thereby aligning the effective memory horizon with the actual pace of distributional change. Moreover, FOCAL incorporates a frequency-domain regularization that captures a wide spectrum of temporal patterns without requiring domain-specific priors. Together, these mechanisms enable accurate, resilient decomposition even when the underlying dynamics are both fast-changing and structurally diverse.

2.2 Coupled Matrix-Tensor Factorization

Previous studies [8, 9, 33, 39, 43] have proposed Coupled Matrix-Tensor Factorization (CMTF) methods to jointly decompose tensors and matrices, enabling effective data fusion. Early work on CMTF, such as CMTF-ALS [2], focuses on optimizing decompositions on a single machine, while large-scale methods [17, 18, 44] employ the MapReduce framework to distribute computation across multiple nodes, achieving scalability for massive datasets. When data exhibit irregular dimensions or require specialized modeling, TASTE [3] exploits a PARAFAC2-based CMTF paradigm tailored to Electronic Health Records (EHRs), and C3APTION [10] extends this approach by coupling nonnegative PARAFAC2 and CP models.

Despite their success in static or batch-processing contexts, existing CMTF methods remain ill-suited to streaming scenarios. These approaches rely on full re-factorization whenever new data become available, which incurs prohibitive computational cost. To the best of our knowledge, no prior work offers a dedicated online framework for CMTF that handles streaming inputs efficiently.

In contrast, FOCAL is designed to explicitly handle streaming data under real-time constraints by employing online update strategies that adapt to both single-tensor streaming and matrix-tensor streaming. This enables FOCAL to maintain high decomposition quality, scale to large and fast-evolving datasets, and capture time-varying patterns essential for real-time applications.

3 Proposed Method

We propose FOCAL, an efficient method for online coupled matrix-tensor factorization. We address the following three key challenges:

- (1) **Accurately decomposing time-evolving data.** A majority of real-world datasets exhibit temporal evolution, where the underlying data distribution changes over time. Capturing these shifts accurately is crucial for maintaining the quality of tensor decomposition. To address this, we present an adaptive forgetting factor mechanism that detects and adapts to temporal changes in the data. This mechanism dynamically adjusts the weighting of past and recent data, emphasizing more recent information for better alignment with the current characteristics of the dataset (Section 3.2).
- (2) **Enhancing the accuracy of temporal factors.** Temporal factors play a critical role in tensor decomposition, as they encapsulate the evolution of data over time. However, these factors are particularly prone to noise and overfitting, which degrade the quality of the decomposition and destabilize the optimization process. To address this, we propose a frequency-domain regularization that exploits the spectral profile of streaming data to enforce coherent trajectories in the time mode. This yields more accurate and noise-resilient temporal factors, which is essential for real-time applications (Section 3.3).

- (3) **Efficiently updating streaming matrix-tensor data.** Traditional methods typically require full retraining when new data become available, which is computationally prohibitive and inefficient. To overcome this, FOCAL reuses factor matrices obtained from prior decomposition steps. By distinguishing between already decomposed factors and newly arrived data, unnecessary computations are avoided, significantly reducing processing time without compromising accuracy. This approach ensures scalability and efficiency, making it particularly well-suited for streaming tensor data (Section 3.4).

3.1 Problem Formulation

We formulate our problem under two distinct settings: single-tensor streaming and matrix-tensor streaming.

Problem 1 (Single-Tensor Streaming).

- Given** (1) Pre-existing factor matrices $A \in \mathbb{R}^{I \times R}$, $B \in \mathbb{R}^{J \times R}$, and $C_{old} \in \mathbb{R}^{K_{old} \times R}$ of old tensor $\mathcal{X}_{old} \in \mathbb{R}^{I \times J \times K_{old}}$
 (2) Pre-existing factor matrices $A \in \mathbb{R}^{I \times R}$ and $D \in \mathbb{R}^{L \times R}$ of coupled matrix $Y \in \mathbb{R}^{I \times L}$
 (3) Newly arrived tensor $\mathcal{X}_{new} \in \mathbb{R}^{I \times J \times K_{new}}$

Compute efficiently the factor matrices of the accumulated tensor $\mathcal{X} = [\mathcal{X}_{old}, \mathcal{X}_{new}]$ and the coupled matrix Y , yielding the new temporal factor $C_{new} \in \mathbb{R}^{K_{new} \times R}$ with $C = [C_{old}, C_{new}]$. $\square$

Problem 2 (Matrix-Tensor Streaming).

- Given** (1) Pre-existing factor matrices $A \in \mathbb{R}^{I \times R}$, $B \in \mathbb{R}^{J \times R}$, and $C_{old} \in \mathbb{R}^{K_{old} \times R}$ of old tensor $\mathcal{X}_{old} \in \mathbb{R}^{I \times J \times K_{old}}$
 (2) Pre-existing factor matrices $C_{old} \in \mathbb{R}^{K_{old} \times R}$ and $D \in \mathbb{R}^{L \times R}$ of old coupled matrix $Y_{old} \in \mathbb{R}^{K_{old} \times L}$
 (3) Newly arrived tensor $\mathcal{X}_{new} \in \mathbb{R}^{I \times J \times K_{new}}$
 (4) Newly arrived coupled matrix $Y_{new} \in \mathbb{R}^{K_{new} \times L}$

Compute efficiently the factor matrices of the accumulated tensor $\mathcal{X} = [\mathcal{X}_{old}, \mathcal{X}_{new}]$ and the coupled matrix $Y = [Y_{old}, Y_{new}]$, yielding $C_{new} \in \mathbb{R}^{K_{new} \times R}$ with $C = [C_{old}, C_{new}]$. $\square$

3.2 Adaptive Objective Function

We start by elaborating a loss function for FOCAL that effectively adapts to temporal changes. Our main approaches are as follows.

- (1) **Distinguishing between old and new data.** By explicitly identifying and treating old and new data differently within the loss function, FOCAL prioritizes computational resources efficiently.
- (2) **Incorporating an adaptive forgetting factor.** To handle temporal evolution in streaming data, we integrate an adaptive forgetting factor into the loss function. This mechanism adjusts the relative importance of historical and recent data, enabling the model to adapt dynamically to changes over time.

For Problem 1, we define the loss function $\mathcal{L}_{ST}$ for single-tensor streaming as follows:

$$\begin{aligned} \mathcal{L}_{ST} = & \lambda \|\mathcal{X}_{old} - \llbracket A, B, C_{old} \rrbracket\|_F^2 \\ & + \|\mathcal{X}_{new} - \llbracket A, B, C_{new} \rrbracket\|_F^2 + \|Y - AD^\top\|_F^2, \end{aligned} \quad (1)$$

where $0 \leq \lambda \leq 1$ is a forgetting factor for the old tensor $\mathcal{X}_{old}$. A smaller value of λ results in the model placing greater emphasis on

fitting more recent data. For Problem 2, the loss function $\mathcal{L}_{MT}$ for matrix-tensor streaming is defined as follows:

$$\mathcal{L}_{MT} = \lambda \|\mathcal{X}_{old} - \llbracket A, B, C_{old} \rrbracket\|_F^2 + \|\mathcal{X}_{new} - \llbracket A, B, C_{new} \rrbracket\|_F^2 + \mu \|Y_{old} - C_{old} D^\top\|_F^2 + \|Y_{new} - C_{new} D^\top\|_F^2, \quad (2)$$

where $0 \leq \lambda, \mu \leq 1$ are forgetting factors for the old tensor $\mathcal{X}_{old}$ and the old matrix Y_{old} , respectively.

In addition to defining the forgetting factors for historical data, FOCAL adaptively adjusts these parameters in real time by monitoring a local error at each streaming step. Specifically, when the factor matrices are updated, the normalized local error e_{loc} is measured via the Frobenius norm between the newly arrived data and its corresponding reconstructed portion. Simultaneously, FOCAL tracks an accumulated average of these local errors, $\bar{e}_{loc}$, over past updates. If the current local error exceeds $\bar{e}_{loc}$, it indicates that the model is underfitting the latest data and should place less emphasis on older data. Conversely, when the local error is smaller than the average, the model can afford to retain more historical information.

To formalize this process, let $\eta \geq 0$ be a learning rate that controls how quickly λ and μ adapt to changing conditions. At each update step, λ and μ are adjusted based on the local errors:

$$\lambda \leftarrow \lambda + \eta (\bar{e}_{loc}^{(tensor)} - e_{loc}^{(tensor)}), \quad \mu \leftarrow \mu + \eta (\bar{e}_{loc}^{(matrix)} - e_{loc}^{(matrix)}).$$

Note that if $e_{loc} > \bar{e}_{loc}$, then the forgetting factor is decreased, thereby reducing the influence of historical data, and vice versa. Moreover, the parameter η has a clear role: if $\eta = 0$, the forgetting factor remains fixed; as η increases, the algorithm becomes more responsive to new data, providing a principled trade-off between historical memory and responsiveness to contemporary patterns.

This adaptive strategy confers three key advantages. First, it *learns* drift-aware weighting directly from the data, eliminating labor-intensive grid search for an optimal decay rate. Second, it employs one-shot closed-form updates that recycle historical factors rather than discarding them, thereby avoiding costly reinitialization when distributional shifts occur. Finally, it provides a unified objective that simultaneously accommodates tensor and coupled-matrix streams without bespoke tuning for each modality. Empirically, this approach achieves consistently lower reconstruction error on standard non-stationary benchmarks (Section 4.4).

3.3 Frequency Regularization

Temporal tensors often exhibit essential patterns that reflect distinct dynamics, such as (i) global trends, (ii) local smoothness, and (iii) seasonality (see Figure 2). These patterns capture key temporal characteristics: overarching shifts over time, gradual changes in the data, and periodic behaviors, respectively. Effectively modeling these dynamics is crucial for accurate tensor decomposition.

We propose a frequency-regularization loss $\mathcal{L}_f$, designed to exploit the relationship between temporal and frequency domains:

$$\mathcal{L}_f = \|\Phi * (FC_{new})\|_F^2, \quad (3)$$

where $\Phi \in \mathbb{R}^{K_{new} \times R}$ is a frequency filter matrix, $F \in \mathbb{C}^{K_{new} \times K_{new}}$ is the Fourier matrix defined as $F = (e^{-2\pi i m n / K_{new}})_{m,n=0, \dots, K_{new}-1}$, and $*$ denotes the Hadamard product. The frequency regularization loss is inspired by the natural correspondences between temporal patterns and distinct frequency characteristics:

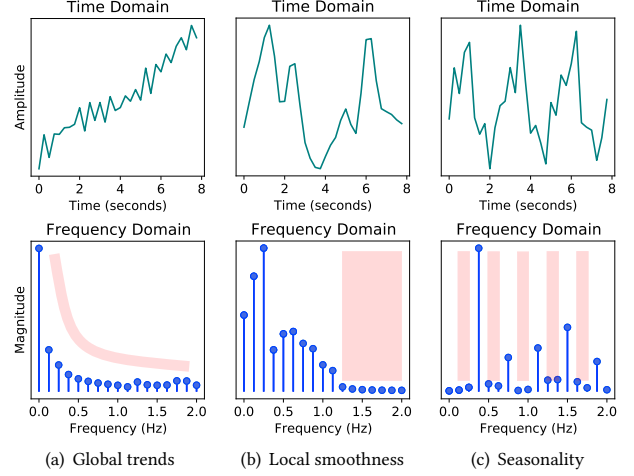


Figure 2: Three representative patterns commonly observed in temporal data and their frequency-domain magnitudes. (a) Global trends correspond to dominant low frequencies, (b) Local smoothness is associated with reduced high-frequency components, and (c) Seasonality is marked by periodic peaks at specific frequencies. By leveraging these distinct frequency characteristics, FOCAL robustly captures detailed patterns.

- (i) *Global trends* align with low-frequency dominance, emphasizing gradual, long-term changes (Figure 2(a)).
- (ii) *Local smoothness* corresponds to high-frequency suppression, as abrupt changes manifest in higher frequencies (Figure 2(b)).
- (iii) *Seasonality* relates to periodic peaks in the frequency domain, capturing regular, repeating patterns (Figure 2(c)).

By incorporating such time-frequency correspondences into the decomposition process, our frequency regularization encourages the temporal factors to align with these natural patterns. For example, consider a frequency filter Φ that is non-zero only in the high-frequency region. Then, minimizing $\mathcal{L}_f$ effectively suppresses the high frequencies of the temporal factor C_{new} , thereby promoting local smoothness. Similarly, by appropriately configuring Φ to emphasize low-frequency dominance or periodic peaks, the properties of global trends and seasonality are achieved, respectively.

To operationalize the filter, we assume that

$$\Phi = [\phi, \dots, \phi] \in \mathbb{R}^{K_{new} \times R}, \quad \phi \in \mathbb{R}^{K_{new}},$$

so the same one-dimensional frequency mask ϕ is shared across all latent components. This *global mask* is well-motivated since most real-world streams exhibit common spectral structure across factors. It also improves identifiability: per-component masks introduce a multiplicative degree of freedom, whereas a shared mask fixes the spectral gauge and stabilizes updates. Importantly, expressivity is preserved—the mask only shapes the admissible frequency band, while component-specific dynamics are captured by the temporal factor C_{new} and its interactions with the non-temporal factors.

When domain knowledge is available, ϕ can be supplied manually. In the general streaming case, however, the temporal pattern of the newly arriving slices is unknown a priori; therefore, we let the model update ϕ adaptively from the data itself.

Let X_{new} (tensor) and Y_{new} (coupled matrix) denote the data that arrive at the current time step. We maintain two running spectra, $\phi_X, \phi_Y \in \mathbb{R}^{K_{new}}$, which are aggregated power-spectra estimates of the tensor and matrix streams, respectively. For each batch, we compute the magnitude of the discrete Fourier transform along the temporal mode, average over the non-temporal modes, and update

$$\begin{aligned}\phi_X &\leftarrow \lambda \phi_X + \frac{1}{IJ} \sum_{i,j} |\mathcal{F}_3(X_{new})| [i, j, :], \\ \phi_Y &\leftarrow \mu \phi_Y + \frac{1}{L} \sum_{\ell} |\mathcal{F}_1(Y_{new})|[:, \ell],\end{aligned}\quad (4)$$

where $\mathcal{F}_n(\cdot)$ denotes a 1-D fast Fourier transform (FFT) along mode n , and $\lambda, \mu \in [0, 1]$ are forgetting factors identical to those used in the loss. When only the tensor streams are present (Setting 1), ϕ_Y is set to zero. The final mask is then defined as

$$\phi = \exp \left(-\frac{\phi_X}{\|\phi_X\|_2} - \frac{\phi_Y}{\|\phi_Y\|_2} \right) \in \mathbb{R}^{K_{new}}.$$

This exponential kernel assigns smaller penalties to frequency bins with consistently high energy in the incoming data and imposes heavier penalties on rarely activated, presumably noisy, frequencies. Because the update in (4) scales only as $O(K_{new} \log K_{new})$ (FFT on the incoming slice), ϕ can be refreshed with minimal overhead. This approach allows the frequency filter to be flexibly adjusted to align with the specific characteristics of each dataset, providing a significant advantage in capturing diverse temporal dynamics.

3.4 Efficient Updates of Factor Matrices

Traditional tensor decomposition methods incur substantial costs by retraining the entire model whenever new data are introduced, which is inefficient for continuous or large-scale settings. FOCAL addresses this by leveraging factor matrices from earlier decomposition steps, updating only the components relevant to the incoming data. This selective strategy eliminates unnecessary recomputation and ensures both computational efficiency and model fidelity.

We derive the update formulations for both Problem 1 (single-tensor streaming) and Problem 2 (matrix-tensor streaming) settings, demonstrating how FOCAL ensures scalability and efficiency in handling streaming tensor data.

3.4.1 Updates for Single-Tensor Streaming. For single-tensor streaming, the loss function is the sum of Equations (1) and (3):

$$\begin{aligned}\mathcal{L} &= \lambda \|X_{old} - \llbracket A, B, C_{old} \rrbracket\|_F^2 + \|X_{new} - \llbracket A, B, C_{new} \rrbracket\|_F^2 \\ &\quad + \|Y - AD^\top\|_F^2 + \|\Phi * (FC_{new})\|_F^2.\end{aligned}\quad (5)$$

Initialization. Before initiating the update process, we first compute the factor matrices A, B, C_{old} , and D from the initial tensor X_{old} and the coupled matrix Y . Additionally, we derive auxiliary matrices $P_1 \in \mathbb{R}^{I \times R}$, $P_2 \in \mathbb{R}^{J \times R}$, and $Q_1, Q_2 \in \mathbb{R}^{R \times R}$ to enable efficient updates in subsequent steps. Following [2], we initialize A, B, C_{old} , and D using the CMTF-ALS method. The auxiliary matrices are then obtained as follows:

$$\begin{aligned}P_1 &\leftarrow X_{old;(1)}(C_{old} \odot B), \quad Q_1 \leftarrow C_{old}^\top C_{old} * B^\top B, \\ P_2 &\leftarrow X_{old;(2)}(C_{old} \odot A), \quad Q_2 \leftarrow C_{old}^\top C_{old} * A^\top A,\end{aligned}$$

where $T_{(n)} \in \mathbb{R}^{I_n \times \prod_{i \neq n} I_i}$ denotes the mode- n unfolding of a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times \dots \times I_N}$, and $\odot$ is the Khatri-Rao product.

Update of the Temporal Factor. When a new data tensor X_{new} is provided, the temporal factor matrix C is updated first, as it influences the updates of all other factor matrices. Because $C = [C_{old}, C_{new}]$, it suffices to compute the new temporal factor matrix C_{new} . Using the loss function in Equation (5), we set $\partial \mathcal{L} / \partial C_{new} = 0$ and derive the following equation for the computation of C_{new} :

LEMMA 1. Solving the following Sylvester equation [7] yields the new temporal factor matrix C_{new} :

$$C_{new}(B^\top B * A^\top A) + \Re(F^H \Psi^2 F) C_{new} = X_{new;(3)}(B \odot A),$$

where F^H denotes the conjugate transpose of F , $\Re(Z)$ is the real part of a complex matrix Z , and $\Psi := \text{Diag}(\phi) \in \mathbb{R}^{K_{new} \times K_{new}}$. That is, Ψ is a diagonal matrix defined as $\Psi_{kk} = \phi_k$ for $k = 0, \dots, K_{new} - 1$. $\square$

PROOF. See Appendix A.1. $\square$

Note that a naïve computation of the matrix $F^H \Psi^2 F$ requires $O(K_{new}^3)$ operations. However, this computation can be significantly accelerated by exploiting the Fourier structure. Indeed,

$$\begin{aligned}(F^H \Psi^2 F)_{m,n} &= \sum_{k=0}^{K_{new}-1} e^{2\pi i m k / K_{new}} \cdot \Psi_{kk}^2 \cdot e^{-2\pi i k n / K_{new}} \\ &= \sum_{k=0}^{K_{new}-1} \phi_k^2 \cdot e^{-2\pi i k (n-m) / K_{new}} = \mathcal{F}[\phi^2]_{n-m},\end{aligned}$$

where $\mathcal{F}[\phi^2]$ denotes the discrete Fourier transform (DFT) of ϕ^2 . Hence, every entry depends only on the index difference $(n - m)$ and is determined by a single DFT of ϕ^2 . Computing this DFT via the FFT reduces the cost from $O(K_{new}^3)$ to $O(K_{new} \log K_{new})$.

Updates of the Non-temporal Factors. By carrying out the same calculus for the remaining non-temporal factor matrices, we obtain the following update rules for A, B , and D :

LEMMA 2. The non-temporal factor matrices are updated as

$$A \leftarrow (P_1 + YD)(Q_1 + D^\top D)^{-1}, \quad B \leftarrow P_2(Q_2)^{-1}, \quad D \leftarrow (A^\dagger Y)^\top,$$

where the auxiliary matrices are computed by

$$\begin{aligned}P_1 &\leftarrow \lambda P_1 + X_{new;(1)}(C_{new} \odot B), \quad Q_1 \leftarrow \lambda Q_1 + C_{new}^\top C_{new} * B^\top B, \\ P_2 &\leftarrow \lambda P_2 + X_{new;(2)}(C_{new} \odot A), \quad Q_2 \leftarrow \lambda Q_2 + C_{new}^\top C_{new} * A^\top A,\end{aligned}$$

and $A^\dagger$ denotes the Moore–Penrose pseudoinverse of A . $\square$

PROOF. See Appendix A.2. $\square$

These update rules are computationally efficient because they leverage the auxiliary matrices P_1, P_2 and Q_1, Q_2 , which are precomputed and incrementally updated from previous steps. For example, recomputing $P_1 = X_{acc;(1)}(C \odot B)$ from scratch at each update requires $O(IJK_{acc}R)$ time, where $X_{acc} \in \mathbb{R}^{I \times J \times K_{acc}}$ is the accumulated tensor along the temporal mode. In contrast, our proposed update requires only $O(IJK_{new}R)$ operation, with $K_{new} \ll K_{acc}$ as the tensor grows. This leads to significantly reduced computational cost, especially in long streaming sequences.

Complexity. We provide the time and space complexity of FOCAL for newly arrived data in the single-tensor streaming setting:

THEOREM 3. For single-tensor streaming, the time complexity and space complexity of FOCAL are given by $O(IJK_{new}R + ILR)$ and $O(IJK_{new} + IL + LR)$, respectively. $\square$

PROOF. See Appendix A.3. $\square$

3.4.2 Updates for Matrix-Tensor Streaming. In the matrix-tensor streaming setting, both the tensor $\mathcal{X}$ and the coupled matrix Y grow over time, reflecting the simultaneous streaming nature of the data. The loss function for matrix-tensor streaming is defined as the sum of Equations (2) and (3):

$$\mathcal{L} = \lambda \|\mathcal{X}_{old} - \llbracket A, B, C_{old} \rrbracket\|_F^2 + \|\mathcal{X}_{new} - \llbracket A, B, C_{new} \rrbracket\|_F^2 \\ + \mu \|Y_{old} - C_{old}D^\top\|_F^2 + \|Y_{new} - C_{new}D^\top\|_F^2 + \|\Phi * (FC_{new})\|_F^2.$$

Initialization. The factor matrices A, B, C_{old} , and D are derived from the initial tensor $\mathcal{X}_{old}$ and the coupled matrix Y_{old} . After initializing the factors using the CMTF-ALS method, the auxiliary matrices are calculated as follows:

$$P_1 \leftarrow X_{old;(1)}(C_{old} \odot B), \quad Q_1 \leftarrow C_{old}^\top C_{old} * B^\top B, \\ P_2 \leftarrow X_{old;(2)}(C_{old} \odot A), \quad Q_2 \leftarrow C_{old}^\top C_{old} * A^\top A, \\ P_3 \leftarrow Y_{old}^\top C_{old}, \quad Q_3 \leftarrow C_{old}^\top C_{old}.$$

Update Rules. We follow exactly the same strategy as in the single-tensor case so that, with minor algebraic modifications, the resulting update rules are obtained in a manner analogous to the derivations given above. The proofs are provided in Appendices A.4-A.5.

LEMMA 4. *The temporal factor C_{new} is computed by solving the following Sylvester equation:*

$$C_{new}(B^\top B * A^\top A + D^\top D) + \Re(F^H \Psi^2 F)C_{new} \\ = X_{new;(3)}(B \odot A) + Y_{new}D,$$

where $\Psi := \text{Diag}(\phi) \in \mathbb{R}^{K_{new} \times K_{new}}$. $\square$

LEMMA 5. *The non-temporal factor matrices are updated as*

$$A \leftarrow P_1(Q_1)^{-1}, \quad B \leftarrow P_2(Q_2)^{-1}, \quad D \leftarrow P_3(Q_3)^{-1},$$

where the auxiliary matrices are computed by

$$P_1 \leftarrow \lambda P_1 + X_{new;(1)}(C_{new} \odot B), \quad Q_1 \leftarrow \lambda Q_1 + C_{new}^\top C_{new} * B^\top B, \\ P_2 \leftarrow \lambda P_2 + X_{new;(2)}(C_{new} \odot A), \quad Q_2 \leftarrow \lambda Q_2 + C_{new}^\top C_{new} * A^\top A, \\ P_3 \leftarrow \mu P_3 + Y_{new}^\top C_{new}, \quad Q_3 \leftarrow \mu Q_3 + C_{new}^\top C_{new}. \quad \square$$

Complexity. The time and space complexity of FOCAL for newly arrived data in the matrix-tensor streaming setting is as follows:

THEOREM 6. *The time complexity and space complexity of FOCAL for matrix-tensor streaming are given by $O(IJK_{new}R + (K_{new} + R)LR)$ and $O(IJK_{new} + K_{new}L + LR)$, respectively.* $\square$

PROOF. See Appendix A.6. $\square$

4 Experiments

We aim to answer the following questions:

- Q1 **Performance (Section 4.2).** How accurately and efficiently does FOCAL update factors compared to baseline methods?
- Q2 **Scalability (Section 4.3).** How does FOCAL scale with increasing sizes of newly arrived data?
- Q3 **Ablation Study (Section 4.4).** How does the adaptive forgetting factor, as well as the frequency regularization, affect the performance of FOCAL?
- Q4 **Anomaly Detection (Section 4.5).** Does FOCAL successfully detect anomalies in streaming data environments?
- Q5 **Visualization (Section 4.6).** Which temporal patterns are revealed through visualization of the learned frequency filter?

Table 2: Dataset summarization.

Dataset	Stream Type	Tensor Size	Matrix Size
ML-100K ³	Single-Tensor	1682 × 943 × 200	1682 × 19
Amazon ⁴	Single-Tensor	901 × 1720 × 200	901 × 170
Stock-US ¹	Matrix-Tensor	500 × 88 × 3651	3651 × 10
Stock-JPN ¹	Matrix-Tensor	300 × 88 × 3856	3856 × 10
Stock-CHN ¹	Matrix-Tensor	1471 × 88 × 1170	1170 × 10

4.1 Experiment Settings

Datasets. We evaluate FOCAL on five real-world datasets covering both single-tensor and matrix-tensor stream types, as described in Table 2. ML-100K contains a *(movie) × (user) × (time)* rating tensor with values from 1 to 5 and a *(movie) × (genre)* matrix constructed by the GroupLens research group. Amazon consists of *(item) × (user) × (time)* binary tensors indicating whether a user wrote a review (1) or not (0), along with a *(item) × (category)* matrix. The original timestamps from ML-100K and Amazon are binned into 200 uniform intervals. In the Stock-US, Stock-JPN, and Stock-CHN datasets¹, both a tensor of *(stock) × (feature) × (day)* and a matrix of *(day) × (feature)* grow over time. The tensor includes five basic stock price features and 83 technical indicators derived from them using the Technical Analysis library². Additionally, the matrix represents global market indicators, including market volatility, interest rates, and macroeconomic trends. We normalize real-world datasets based on their characteristics. For stock datasets, we apply z-normalization to each column representing stock features to ensure a consistent scale.

Baselines. We compare our FOCAL against six baselines:

- (1) **OnlineCP**⁵ [54]: a streaming tensor decomposition method, which does not incorporate matrix coupling or regularization.
- (2) **TeCPSGD**⁶ [36]: a gradient-descent-based method for streaming tensors using subspace learning.
- (3) **SOAP**⁷ [38]: a second-order optimization based CP method in online tasks, but is limited in handling coupled matrix updates.
- (4) **SOFA**⁸ [29]: a seasonality-aware online CP factorization method via temporal-pattern detection.
- (5) **ROLCP**⁹ [1]: a randomized online CP method for efficient computation using a fixed forgetting factor.
- (6) **CMTF-ALS**¹ [2]: an ALS-based algorithm that re-runs a CMTF update from scratch whenever new data arrive.

Implementation and Hyperparameters. Our system utilizes an Intel Core i7-10700KF @ 3.80GHz processor paired with 32GB of RAM. All the methods are implemented in Matlab. Prior to the update cycle, we initialize the factor matrices so that the initialization phase spans 10% of the total period. For each update cycle, newly arriving data have a batch size of $K_{new} = 10$ for single-tensor streaming and $K_{new} = 20$ for matrix-tensor streaming. We fix the rank $R = 50$ (Single-tensor) or $R = 10$ (Matrix-tensor), and set the

¹<https://github.com/snudatalab/FOCAL/>

²<https://technical-analysis-library-in-python.readthedocs.io/>

³<https://grouplens.org/datasets/movielens/100k/>

⁴<https://www.kaggle.com/datasets/deovcs/amazon-dataset>

⁵<https://shuozhou.github.io/>

⁶<https://github.com/hirokyu-kasai/OLSTEC>

⁷<https://drive.google.com/drive/folders/1x6PdEsr-1xDccm7titi5dQPLwgcKuOii>

⁸<https://github.com/wooner49/sofia>

⁹<https://github.com/thanhtbt/ROLCP>

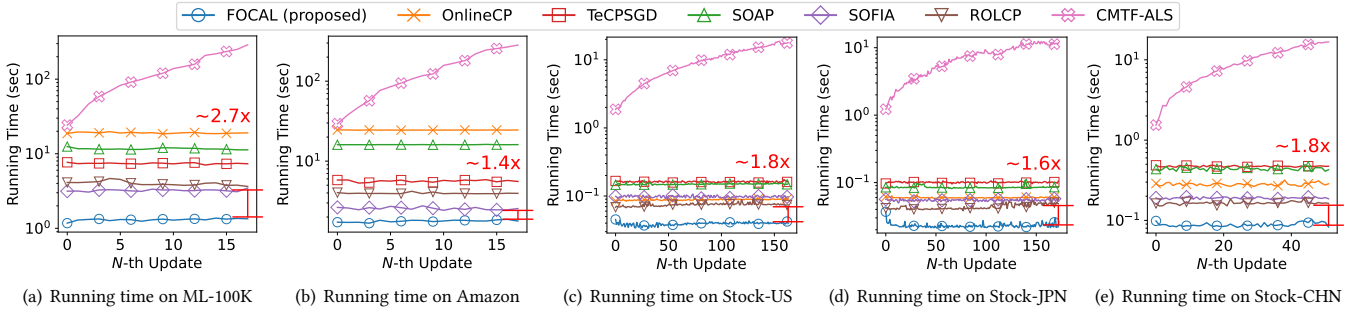


Figure 3: Running time of FOCAL vs. baselines on various datasets. (a)–(b) show single-tensor streaming results while (c)–(e) depict matrix-tensor streaming results. Each method is represented by a line connecting all update points, with a few updates highlighted by large markers. FOCAL consistently outperforms other methods with up to $2.7\times$ faster speed.

maximum number of iterations to 10, following common practice in prior work to ensure fair and consistent comparison. The spectra ϕ_X and ϕ_Y are initialized to zero. We set the initial forgetting factors to $\lambda, \mu = 0.9$ and adaptively update them with a learning rate of $\eta = 0.02$, keeping them in $[0.8, 1.0]$ via clipping. Note that the clipping-based update rule is an efficient way to keep the forgetting factor in the range without adding computational overhead—a critical benefit in streaming scenarios. For the gradient-descent-based method (TeCPSGD), we adopt the optimal hyperparameters recommended in the original paper to ensure competitive performance.

Evaluation Metrics. We measure both *local* and *global* reconstruction errors. Because some of the baselines do not support decomposing a coupled matrix, we measure only the reconstruction error of the primary tensor for all approaches to ensure a fair comparison. Given a tensor X , denote its reconstruction from factor matrices as $\hat{X}$. Suppose X_{new} is the newly arrived tensor slice at a particular update and X_{acc} is the accumulated tensor (including all past and newly arrived slices). We define the local and global errors as follows: $E_{local} = \frac{\|X_{new} - \hat{X}_{new}\|_F}{\|X_{new}\|_F}$ and $E_{global} = \frac{\|X_{acc} - \hat{X}_{acc}\|_F}{\|X_{acc}\|_F}$.

4.2 Performance (Q1)

We compare FOCAL against six baseline methods. Figure 3 presents the per-update speed comparisons, and Table 3 provides a quantitative summary of local errors across different datasets.

As seen in Figure 3, FOCAL consistently achieves the lowest running time, demonstrating its capability to adapt efficiently to recent data. Compared to the static approach that re-factorizes from scratch (CMTF-ALS), FOCAL’s update time is up to an order of magnitude faster. Furthermore, Table 3 shows that FOCAL achieves significantly lower local error than the second-best competitor, demonstrating its superior accuracy. Although SOFIA and ROLCP offer reasonably quick updates, they yield higher local errors since they rely on fixed forgetting factors, capture only one kind of temporal pattern (seasonality), and ignore the complementary information in the coupled matrix. Overall, FOCAL attains significantly lower errors, and converges more quickly in per-update runtime.

4.3 Scalability (Q2)

We investigate how well FOCAL scales with the amount of newly arrived data. Specifically, we vary the batch size of incoming slices (for single-tensor streaming) or the batch size of new slices and

new matrix rows (for matrix-tensor streaming). Figure 4 shows the running times on single-tensor and matrix-tensor streaming as a function of batch size K_{new} . We present our results using box plots, where the orange line represents the median. The box is defined by the first and third quartiles ($Q1$ and $Q3$), while two horizontal lines correspond to $2.5Q3 - 1.5Q1$ and $2.5Q1 + 1.5Q3$. FOCAL exhibits near linear scalability with respect to new data size, ensuring computational efficiency even with large-scale data.

4.4 Ablation Study (Q3)

We investigate how the adaptive forgetting factor and the frequency regularization influence the performance of FOCAL.

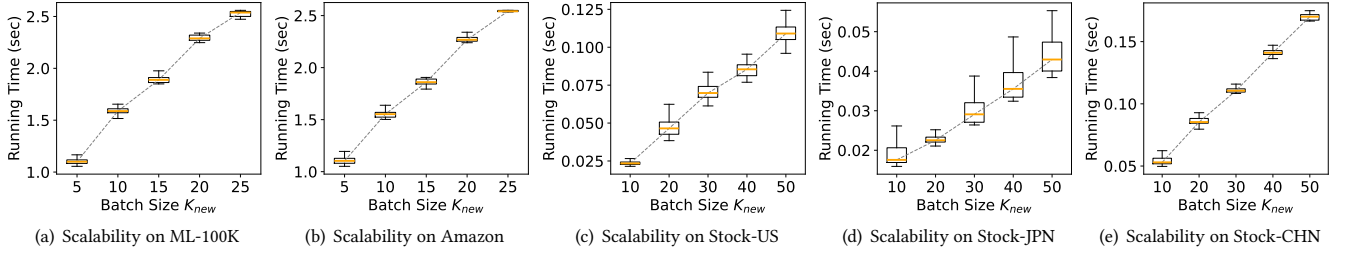
Adaptive Forgetting Factor. We compare *fixed* manual forgetting factors $\lambda, \mu = 0.9$ with our *adaptive* tuning strategy, where FOCAL dynamically adjusts them within $[0.8, 1.0]$ (with an initial value of 0.9). As shown in Table 4, the adaptive strategy consistently reduces local errors with comparable global errors. In a streaming setting, emphasizing local errors is paramount because recent data better capture evolving real-world dynamics. Therefore, capturing these recent trends is more important than minimizing the overall global errors. Our results confirm that adaptively adjusting λ and μ successfully achieves this goal, making FOCAL more effective in real-time streaming scenarios.

Frequency Regularization. We assess the contribution of our frequency-domain regularization by ablating the filter-learning step from FOCAL. Table 5 shows that dynamically estimating the spectral filter introduces only a negligible runtime overhead (under 2% per update), yet consistently reduces local reconstruction error by a substantial margin. Moreover, our qualitative visualizations (Section 4.6) show that the learned frequency filters adapt to dataset-specific seasonality and smoothness, plausibly explaining the observed reductions in local reconstruction error. These results indicate that the data-driven frequency prior steers the optimizer more effectively, yielding better accuracy while leaving update speed essentially unchanged.

Sensitivity to Learning Rate. We perform an ablation study on the adaptive forgetting factor’s learning rate η , evaluating its influence on reconstruction accuracy and assessing how precisely the model tracks the velocity of data drift at each update step. Figure 5 investigates the sensitivity of FOCAL to the learning rate η by reporting the mean local reconstruction error across a wide grid of

Table 3: Local error comparison. We compute the mean and standard deviation across all updates. The lowest and second-lowest local errors are highlighted in bold and underlined, respectively. FOCAL outperforms competitors with the lowest local errors.

Method	Single-Tensor Streaming		Matrix-Tensor Streaming		
	ML-100K	Amazon	Stock-US	Stock-JPN	Stock-CHN
OnlineCP [54]	0.9880 $\pm$ 0.0137	0.9961 $\pm$ 0.0056	0.8607 $\pm$ 0.0822	0.8272 $\pm$ 0.0721	0.7930 $\pm$ 0.0388
TeCPSGD [36]	0.9793 $\pm$ 0.0103	0.9908 $\pm$ 0.0092	0.7719 $\pm$ 0.0996	0.7394 $\pm$ 0.0740	0.7027 $\pm$ 0.0552
SOAP [38]	1.2773 $\pm$ 0.2316	1.6224 $\pm$ 0.6292	0.9700 $\pm$ 0.1842	0.9438 $\pm$ 0.1737	0.9465 $\pm$ 0.1756
SOFIA [29]	0.9925 $\pm$ 0.0165	0.9970 $\pm$ 0.0216	0.7144 $\pm$ 0.1712	0.6687 $\pm$ 0.1631	0.7201 $\pm$ 0.1836
ROLCP [1]	0.9938 $\pm$ 0.0070	0.9980 $\pm$ 0.0016	<u>0.6754 $\pm$ 0.0617</u>	<u>0.6563 $\pm$ 0.0727</u>	<u>0.6793 $\pm$ 0.0845</u>
CMTF-ALS [2]	<u>0.9335 $\pm$ 0.0506</u>	<u>0.9744 $\pm$ 0.0413</u>	0.9628 $\pm$ 0.0807	0.9013 $\pm$ 0.0903	0.8803 $\pm$ 0.0427
FOCAL (proposed)	0.8050 $\pm$ 0.0642	0.9656 $\pm$ 0.0155	0.6044 $\pm$ 0.0663	0.6097 $\pm$ 0.0522	0.5614 $\pm$ 0.0362

**Figure 4: Scalability of FOCAL as the size of newly arrived data increases. Note that the running time of FOCAL grows near linearly with data size, indicating that FOCAL scales efficiently, handling larger data sizes with minimal overhead.****Table 4: Comparison between manual and adaptive forgetting factor. Our adaptive tuning yields improvements in local errors with comparable global errors.**

Dataset	Local Error		Global Error	
	Manual	Adaptive	Manual	Adaptive
ML-100K	0.8218	0.8050	1.6462	1.6681
Amazon	0.9808	0.9656	2.3655	2.4309
Stock-US	0.6293	0.6044	5.5264	6.3022
Stock-JPN	0.6184	0.6097	3.8301	4.2670
Stock-CHN	0.5733	0.5614	1.8545	2.0864

Table 5: Impact of Frequency Regularization (FR). Frequency regularization markedly reduces reconstruction error while incurring only a slight increase in update time.

Dataset	Local Error		Update Time (sec)	
	w/o FR	w/ FR	w/o FR	w/ FR
ML-100K	0.8163	0.8050	1.1246	1.1374
Amazon	0.9727	0.9656	1.7502	1.7763
Stock-US	0.6195	0.6044	0.0389	0.0396
Stock-JPN	0.6212	0.6097	0.0225	0.0230
Stock-CHN	0.5706	0.5614	0.0871	0.0889

values. In every dataset, the error surface exhibits a clear U-shape: extremely small learning rates produce large errors because the forgetting factor cannot adjust quickly enough to accommodate sudden accelerations or decelerations in the stream, whereas excessively large rates yield similarly poor accuracy by causing the algorithm to chase temporary fluctuations that do not reflect the underlying drift. The lowest errors are attained at intermediate values, indicating that robust performance hinges on a careful balance between responsiveness and stability.

Figure 6 sheds light on this phenomenon by plotting the evolution of the forgetting factor for three representative learning rates on Stock-US. When η is very small, the trajectory remains almost flat—essentially mimicking a fixed forgetting factor—so the model under-reacts to genuine changes in data velocity. In contrast, a very large η makes the trajectory oscillate sharply, over-emphasizing the newest observations and amplifying noise. Only the intermediate setting produces smooth, well-damped adjustments that minimize reconstruction error and avoid the instability.

4.5 Anomaly Detection (Q4)

We use FOCAL’s adaptive forgetting factor λ to identify abrupt distribution shifts. We analyze the temporal variation of λ , defined as the absolute difference between consecutive values, $|\lambda^{(t)} - \lambda^{(t-1)}|$. Significant fluctuations in this measure indicate abrupt distribution shifts, enabling effective anomaly detection in streaming data. For Stock-US, Figure 7 marks two updates where the forgetting factor changes substantially. These points align with major market disruptions: (i) COVID-19 market crash and (ii) Fed’s interest rate hike outlook. These findings confirm that FOCAL not only maintains accurate factorization over time but also unveils real-world events or structural changes by exploiting its forgetting factor.

4.6 Visualization (Q5)

We visualize the frequency mask ϕ learned by FOCAL to examine whether characteristic temporal patterns emerge. Figure 8 shows that the filter adaptively emphasizes distinct spectral bands over time, revealing non-stationary, time-varying patterns. Using these filters as data-driven priors in subsequent updates plausibly explains the quantitative gains reported earlier, suggesting that the learned spectral structure helps guide the optimization toward lower local error. See Appendix B for additional experiments.

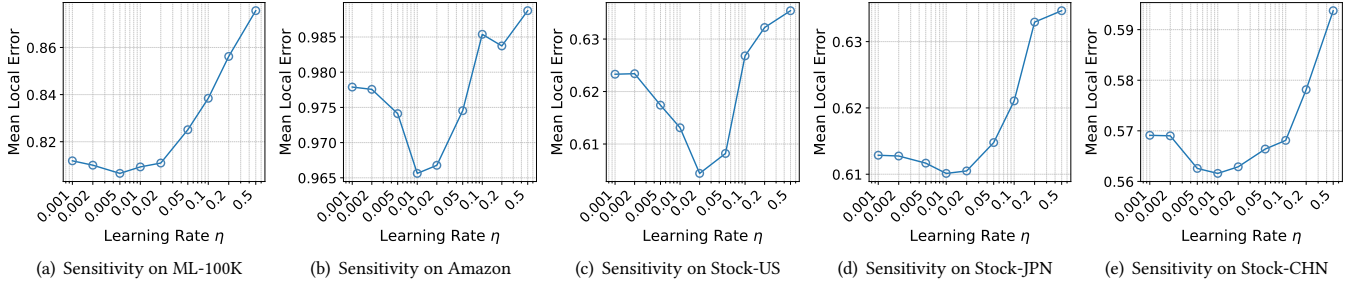


Figure 5: Sensitivity of FOCAL to the learning rate η . Across all datasets, the error curve exhibits a U-shape, implying that an appropriately balanced learning rate is essential for achieving the optimal trade-off between responsiveness and stability.

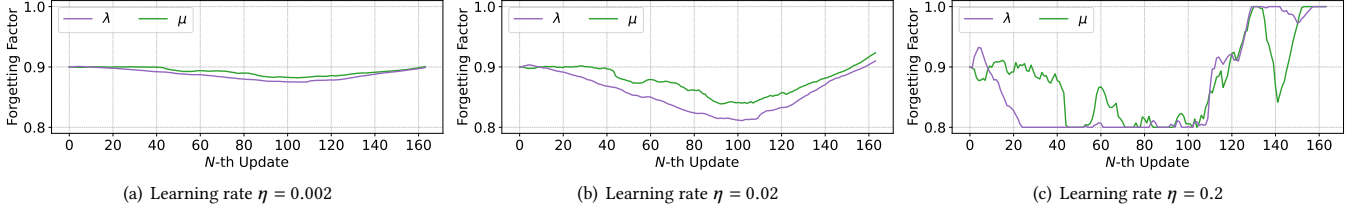


Figure 6: Evolution of the adaptive forgetting factors λ and μ on Stock-US for three learning-rate settings (all curves initialized at 0.9 and clipped to the interval $[0.8, 1.0]$). Empirically, $\eta = 0.02$ yields the sharpest accuracy gains (as seen in Figure 5(c)), which indicates that the associated evolution of the forgetting factors most closely aligns with the data’s underlying drift.

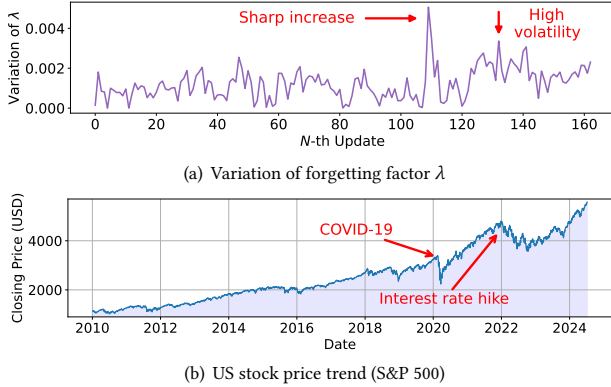


Figure 7: Anomaly detection via FOCAL. Sudden changes in the forgetting factor coincide with abrupt market events.

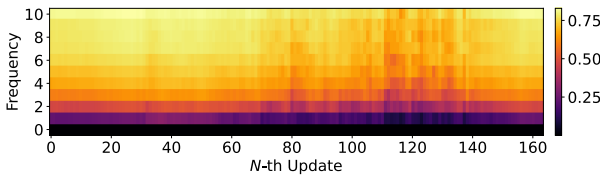


Figure 8: Evolution of ϕ for Stock-US across updates (only half the spectrum is shown since ϕ is symmetric). Low frequencies are consistently weakly penalized, revealing a persistent global trend. High frequencies are strongly down-weighted in updates 0-80, indicating temporal smoothness, whereas updates 110-140 (the COVID-19 era) display relaxed high-frequency penalization, suggesting increased volatility.

5 Conclusion

In this paper, we propose FOCAL, an efficient and scalable method for online CMTF. By integrating an adaptive forgetting factor with a frequency regularization mechanism, FOCAL effectively handles streaming data, accurately capturing temporal dynamics while maintaining computational efficiency. FOCAL distinguishes between old and new data to optimize factor updates, thereby ensuring scalability in real-time applications. Extensive experiments demonstrate FOCAL’s superior performance in terms of both accuracy and speed compared to the state-of-the-art methods. In addition, FOCAL successfully detects the anomalous points in real-world data, highlighting its practical utility for identifying irregular patterns in dynamic environments. As future work, we plan to extend FOCAL to higher-order and sparse data, and systematically develop and evaluate its anomaly detection capability using standardized benchmarks to further validate its effectiveness.

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References

- [1] Karim Abed-Meraim, Nguyen Link Trung, Adel Hafiane, et al. 2021. A fast randomized adaptive CP decomposition for streaming tensors. In *ICASSP*. IEEE.
- [2] Evrim Acar, Tamara G Kolda, and Daniel M Dunlavy. 2011. All-at-once optimization for coupled matrix and tensor factorizations. *arXiv preprint arXiv:1105.3422* (2011).
- [3] Ardavan Afshar, Ioakeim Perros, Haesun Park, Christopher Defilippi, Xiaowei Yan, Walter Stewart, Joyce Ho, and Jimeng Sun. 2020. Taste: temporal and static tensor factorization for phenotyping electronic health records. In *CHIL*.
- [4] Dawon Ahn, Seyun Kim, and U Kang. 2021. Accurate online tensor factorization for temporal tensor streams with missing values. In *CIKM*.
- [5] Miguel Araujo, Pedro Ribeiro, Hyun Ah Song, and Christos Faloutsos. 2019. Tensorcast: forecasting and mining with coupled tensors. *Knowledge and Information Systems* 59, 3 (2019).
- [6] Thirunavukarasu Balasubramaniam, Richi Nayak, and Chau Yuen. 2018. Nonnegative coupled matrix tensor factorization for smart city spatiotemporal pattern mining. In *LOD*. Springer.
- [7] Richard H Bartels and George W Stewart. 1972. Algorithm 432 [C2]: Solution of the matrix equation $AX + XB = C$ [F4]. *Commun. ACM* 15, 9 (1972).
- [8] Alex Beutel, Partha Pratim Talukdar, Abhimanu Kumar, Christos Faloutsos, Evangelos E Papalexakis, and Eric P Xing. 2014. Flexifact: Scalable flexible factorization of coupled tensors on hadoop. In *SDM*. SIAM.
- [9] Dongjin Choi, Jun-Gi Jang, and U Kang. 2019. S3 CMTF: Fast, accurate, and scalable method for incomplete coupled matrix-tensor factorization. *PLoS one* 14, 6 (2019).
- [10] Ekta Guiral, Georgios Theodorou, and Evangelos E Papalexakis. 2020. C 3 APTION: Constrained Coupled CP And PARAFAC2 Tensor Decomposition. In *ASONAM*. IEEE.
- [11] Ekta Gujral, Ravdeep Pasricha, and Evangelos E Papalexakis. 2018. Sambaten: Sampling-based batch incremental tensor decomposition. In *SDM*. SIAM.
- [12] Ekta Gujral, Georgios Theodorou, and Evangelos E Papalexakis. 2020. Spade: Streaming parafac2 decomposition for large datasets. In *SDM*. SIAM.
- [13] Ka Hyun Park, Junhun Kim, and U Kang. 2025. Accurate semi-supervised automatic speech recognition for ordinary and characterized speeches via multi-hypotheses-based curriculum learning. *PLoS One* 20, 10 (2025).
- [14] Jun-Gi Jang and U Kang. 2023. Static and streaming tucker decomposition for dense tensors. *TKDD* 17, 5 (2023).
- [15] Jun-Gi Jang, Jeongyoung Lee, Yong-chan Park, and U Kang. 2023. Fast and Accurate Dual-Way Streaming PARAFAC2 for Irregular Tensors-Algorithm and Application. In *KDD*.
- [16] Jun-Gi Jang, Yong-chan Park, and U Kang. 2024. Fast and accurate parafac2 decomposition for time range queries on irregular tensors. In *CIKM*.
- [17] ByungSoo Jeon, Inah Jeon, Lee Sael, and U Kang. 2016. Scout: Scalable coupled matrix-tensor factorization-algorithm and discoveries. In *ICDE*. IEEE.
- [18] Inah Jeon, Evangelos E Papalexakis, U Kang, and Christos Faloutsos. 2015. Haten2: Billion-scale tensor decompositions. In *ICDE*. IEEE.
- [19] Hiroyuki Kasai. 2016. Online low-rank tensor subspace tracking from incomplete data by CP decomposition using recursive least squares. In *ICASSP*. IEEE.
- [20] Jongjin Kim and U Kang. 2025. Sequentially diversified and accurate recommendations in chronological order for a series of users. In *WSDM*.
- [21] Junhun Kim, Ka Hyun Park, Jun-Gi Jang, and U Kang. 2024. Fast and accurate domain adaptation for irregular tensor decomposition. In *KDD*.
- [22] Junhun Kim, Ka Hyun Park, and U Kang. 2024. Accurate Semi-supervised Automatic Speech Recognition via Multi-hypotheses-Based Curriculum Learning. In *PAKDD*. Springer.
- [23] Junhun Kim, Ka Hyun Park, Hoyoung Yoon, and U Kang. 2024. Domain-Aware Data Selection for Speech Classification via Meta-Reweighting. In *INTERSPEECH*.
- [24] Junhun Kim, Ka Hyun Park, Hoyoung Yoon, and U Kang. 2025. Accurate link prediction for edge-incomplete graphs via PU learning. In *AAAI*, Vol. 39.
- [25] JinGee Kim, Yong-chan Park, Jaemin Hong, and U Kang. 2024. Accurate Stock Movement Prediction via Multi-Scale and Multi-Domain Modeling. In *BigData*. IEEE.
- [26] Junhun Kim, Hoyoung Yoon, Ka Hyun Park, and U Kang. 2025. Accurate graph-based multi-positive unlabeled learning via disentangled multi-view feature propagation. In *KDD*.
- [27] Minjun Kim, Jaehyeon Choi, Seungjoo Lee, Jinhong Jung, and U Kang. 2025. Augward: Augmentation-aware representation learning for accurate graph classification. In *PAKDD*. Springer.
- [28] Taehyung Kwon, Inkyu Park, Dongjin Lee, and Kijung Shin. 2021. Slicenstitch: Continuous cp decomposition of sparse tensor streams. In *ICDE*. IEEE.
- [29] Dongjin Lee and Kijung Shin. 2021. Robust factorization of real-world tensor streams with patterns, missing values, and outliers. In *ICDE*. IEEE.
- [30] Jaeri Lee and U Kang. 2025. Context-aware Sequential Bundle Recommendation via User-specific Representations. In *CIKM*.
- [31] Jaeri Lee, Jeongin Yun, and U Kang. 2024. Towards true multi-interest recommendation: enhanced scheme for balanced interest training. In *BigData*. IEEE.
- [32] Seungjoo Lee, Seokho Ahn, and YoungDuk Seo. 2023. Relation Modeling on Knowledge Graph for Interoperability in Recommender Systems. In *SAC*.
- [33] SeungJoo Lee, Yong-chan Park, and U Kang. 2024. Accurate Coupled Tensor Factorization with Knowledge Graph. In *BigData*. IEEE.
- [34] SeungJoo Lee, Yong-chan Park, and U Kang. 2025. Offline and online coupled tensor factorization with knowledge graph. *PLoS one* 20, 11 (2025).
- [35] Seungjoo Lee, Yong-chan Park, and U Kang. 2025. SwaGNER: Leveraging Span-aware Grid Transformers for Accurate Nested Named Entity Recognition. In *CIKM*.
- [36] Morteza Mardani, Gonzalo Mateos, and Georgios B Giannakis. 2015. Subspace learning and imputation for streaming big data matrices and tensors. *IEEE Transactions on Signal Processing* 63, 10 (2015).
- [37] Mehrnaz Najafi, Lifang He, and S Yu Philip. 2019. Outlier-Robust Multi-Aspect Streaming Tensor Completion and Factorization.. In *IJCAI*.
- [38] Viet-Dung Nguyen, Karim Abed-Meraim, and Nguyen Linh-Trung. 2017. Second-order optimization based adaptive PARAFAC decomposition of three-way tensors. *Digital Signal Processing* 63 (2017).
- [39] Evangelos E Papalexakis, Christos Faloutsos, Tom M Mitchell, Partha Pratim Talukdar, Nicholas D Sidiropoulos, and Brian Murphy. 2014. Turbo-sm: Accelerating coupled sparse matrix-tensor factorizations by 200x. In *SDM*. SIAM.
- [40] Yong-chan Park, Jun-Gi Jang, and U Kang. 2021. Fast and accurate partial fourier transform for time series data. In *KDD*.
- [41] Yong-chan Park, Jongjin Kim, and U Kang. 2024. Fast multidimensional partial fourier transform with automatic hyperparameter selection. In *KDD*.
- [42] Yong-chan Park, Kisoo Kim, and U Kang. 2025. PuzzleTensor: a method-agnostic data transformation for compact tensor factorization. In *KDD*.
- [43] Carla Schenker, Xiulin Wang, and Evrim Acar. 2023. PARAFAC2-based coupled matrix and tensor factorizations. In *ICASSP*. IEEE.
- [44] Kijung Shin, Lee Sael, and U Kang. 2016. Fully scalable methods for distributed tensor factorization. *TKDE* 29, 1 (2016).
- [45] Shaden Smith, Kejun Huang, Nicholas D Sidiropoulos, and George Karypis. 2018. Streaming tensor factorization for infinite data sources. In *SDM*. SIAM.
- [46] Yongseok Soh, Patrick Flick, Xing Liu, Shaden Smith, Fabio Checconi, Fabrizio Petrini, and Jee Choi. 2021. High performance streaming tensor decomposition. In *IPDPS*. IEEE.
- [47] Sangjun Son, Yong-chan Park, Minyong Cho, and U Kang. 2022. DAO-CP: Data-Adaptive Online CP decomposition for tensor stream. *PLoS one* 17, 4 (2022).
- [48] Qingquan Song, Xiao Huang, Hancheng Ge, James Caverlee, and Xia Hu. 2017. Multi-aspect streaming tensor completion. In *KDD*.
- [49] Xiaobing Sun, Jiabao Zhang, Qiming Zhao, Shenghua Liu, Jinglei Chen, Ruoyu Zhuang, Huawei Shen, and Xueqi Cheng. 2021. Cubeflow: Money laundering detection with coupled tensors. In *PAKDD*. Springer.
- [50] Houping Xiao, Fei Wang, Fenglong Ma, and Jing Gao. 2018. eOTD: An efficient online tucker decomposition for higher order tensors. In *ICDM*. IEEE.
- [51] Jeongin Yun, Jaeri Lee, and U Kang. 2025. DART: Diversified and Accurate Long-Tail Recommendation. In *PAKDD*. Springer.
- [52] Zheng Zhang and Cole Hawkins. 2018. Variational Bayesian inference for robust streaming tensor factorization and completion. In *ICDM*. IEEE.
- [53] Shuo Zhou, Sarah Erfani, and James Bailey. 2018. Online CP decomposition for sparse tensors. In *ICDM*. IEEE.
- [54] Shuo Zhou, Nguyen Xuan Vinh, James Bailey, Yunzhe Jia, and Ian Davidson. 2016. Accelerating online cp decompositions for higher order tensors. In *KDD*.

A Proofs

A.1 Proof of Lemma 1

PROOF. We begin by taking the partial derivative of the loss

$$\begin{aligned} \mathcal{L} = & \lambda \|X_{old} - \llbracket A, B, C_{old} \rrbracket\|_F^2 + \|X_{new} - \llbracket A, B, C_{new} \rrbracket\|_F^2 \\ & + \|Y - AD^T\|_F^2 + \|\Phi * (FC_{new})\|_F^2 \end{aligned}$$

with respect to C_{new} and setting it equal to zero.

First, note that the derivative of $\|X_{new} - \llbracket A, B, C_{new} \rrbracket\|_F^2$ with respect to C_{new} can be expressed as

$$\begin{aligned} & \frac{\partial}{\partial C_{new}} \|X_{new;(3)} - C_{new}(B \odot A)^T\|_F^2 \\ & = 2 \left(C_{new}(B \odot A)^T (B \odot A) - X_{new;(3)} (B \odot A) \right) \\ & = 2 \left(C_{new}(B^T B * A^T A) - X_{new;(3)} (B \odot A) \right) \end{aligned}$$

since $(B \odot A)^\top (B \odot A) = B^\top B * A^\top A$. Next, the derivative of $\|\Phi * (FC_{new})\|_F^2$ with respect to C_{new} is given by

$$\frac{\partial}{\partial C_{new}} \|\Phi * (FC_{new})\|_F^2 = F^H (\Phi * \Phi * FC_{new}) + F^\top (\Phi * \Phi * \bar{F}C_{new}).$$

Because Φ and C_{new} are real matrices, we have $\overline{F^\top (\Phi * \Phi * \bar{F}C_{new})} = F^H (\Phi * \Phi * FC_{new})$, which leads to

$$\frac{\partial}{\partial C_{new}} \|\Phi * (FC_{new})\|_F^2 = 2\Re(F^H (\Phi * \Phi * FC_{new})).$$

Collecting these terms and setting the derivative to zero yields

$$C_{new}(B^\top B * A^\top A) + \Re(F^H (\Phi * \Phi * FC_{new})) = X_{new;(3)}(B \odot A).$$

Assuming Φ takes the form $\Phi = [\phi, \dots, \phi] \in \mathbb{R}^{K_{new} \times R}$, where the same filter ϕ is applied to each latent dimension, we have

$$\begin{aligned} & F^H (\Phi * \Phi * FC_{new}) + F^\top (\Phi * \Phi * \bar{F}C_{new}) \\ &= F^H (\Psi \cdot \Psi \cdot FC_{new}) + F^\top (\Psi \cdot \Psi \cdot \bar{F}C_{new}) \\ &= (F^H \Psi^2 F + F^\top \Psi^2 \bar{F})C_{new} \\ &= 2\Re(F^H \Psi^2 F)C_{new}, \end{aligned}$$

where $\Psi = \text{Diag}(\phi)$. Thus, the derivative with respect to C_{new} reduces to a form involving the Sylvester equation

$$C_{new}(B^\top B * A^\top A) + \Re(F^H \Psi^2 F)C_{new} = X_{new;(3)}(B \odot A),$$

hence the desired result in Lemma 1. $\square$

A.2 Proof of Lemma 2

PROOF. The relevant portion of the loss function for A is

$$\lambda \|\mathcal{X}_{old} - \llbracket A, B, C_{old} \rrbracket\|_F^2 + \|\mathcal{X}_{new} - \llbracket A, B, C_{new} \rrbracket\|_F^2 + \|Y - AD^\top\|_F^2.$$

By taking the derivative with respect to A and setting it to zero, we arrive at the following equation:

$$\begin{aligned} & \lambda(P_{1;old}) + X_{new;(1)}(C_{new} \odot B) \\ & - A[\lambda(Q_{1;old}) + (C_{new}^\top C_{new} * B^\top B)] + YD - A(D^\top D) = 0. \end{aligned}$$

Rearranging terms yields the closed-form update

$$A \leftarrow (P_{1;new} + YD)(Q_{1;new} + D^\top D)^{-1},$$

where $P_{1;new}$ and $Q_{1;new}$ incorporate both the old and new parts.

Similar to the update for A , we focus on the portions of the objective function that depend on B :

$$\lambda \|\mathcal{X}_{old} - \llbracket A, B, C_{old} \rrbracket\|_F^2 + \|\mathcal{X}_{new} - \llbracket A, B, C_{new} \rrbracket\|_F^2.$$

Taking the partial derivative with respect to B and setting it to zero leads to $\lambda(P_{2;old}) + X_{new;(2)}(C_{new} \odot A) - B[\lambda(Q_{2;old}) + (C_{new}^\top C_{new} * A^\top A)] = 0$. Solving for B yields

$$B \leftarrow P_{2;new}(Q_{2;new})^{-1},$$

where $P_{2;new}$ and $Q_{2;new}$ are the updated auxiliary matrices.

Finally, recall that the coupling to the matrix Y is reflected in the term $\|Y - AD^\top\|_F^2$. Taking the derivative of this term with respect to D and setting it to zero gives $-2A^\top(Y - AD^\top) = 0$, or $A^\top Y = A^\top AD^\top$. Provided A has full column rank, the Moore-Penrose pseudoinverse $A^\dagger = (A^\top A)^{-1}A^\top$ exists, and hence

$$D = (A^\dagger Y)^\top.$$

This completes the proof of Lemma 2. $\square$

A.3 Proof of Theorem 3

PROOF. The time complexity of FOCAL is proportional to the summation of the computational costs for updating the factor matrices A, B, C , and D . Computing the terms in Lemma 1 requires $O(IJK_{new}R + (I + J + K_{new})R^2 + K_{new}^2R)$, and solving the Sylvester equation takes $O((R + K_{new})^3)$ operations [7]. If we assume that K_{new} and R are sufficiently small compared to I and J , the dominant term is $O(IJK_{new}R)$ for updating C . Next, updating A requires $O(IJK_{new}R + ILR)$ costs due to Lemma 2, and similarly, updating B takes $O(IJK_{new}R)$ time. The update of D involves computing $A^\dagger$ and multiplying $A^\dagger Y$, which take $O(IR^2)$ and $O(ILR)$ operations, respectively. Summing up the computational costs of the four factor matrices, the total time complexity is $O(IJK_{new}R + ILR)$.

For the space complexity, we store the factor matrices A, B, C_{new} , and D , which requires a total of $R(I + J + K_{new} + L)$ space. The newly arrived tensor $\mathcal{X}_{new}$ and the coupled matrix Y require IJK_{new} and IL spaces, respectively. We need $O(IJK_{new} + R(I + J + K_{new} + R) + (I + J)K_{new}R)$ space to compute the auxiliary matrices P_1, P_2 and Q_1, Q_2 . With $K_{new}, R \ll I, J$, these lead to a total of $O(IJK_{new} + IL + LR)$ space complexity. $\square$

A.4 Proof of Lemma 4

PROOF. For matrix-tensor streaming, the relevant loss function portion for updating C_{new} is

$$\|\mathcal{X}_{new} - \llbracket A, B, C_{new} \rrbracket\|_F^2 + \|Y_{new} - C_{new}D^\top\|_F^2 + \|\Phi * (FC_{new})\|_F^2.$$

Taking the derivative with respect to C_{new} and following the proof of A.1, we obtain

$$\begin{aligned} & C_{new}(B^\top B * A^\top A + D^\top D) + \Re(F^H \Psi^2 F)C_{new} \\ &= X_{new;(3)}(B \odot A) + Y_{new}D. \end{aligned}$$

This is precisely the Sylvester equation stated in Lemma 4. $\square$

A.5 Proof of Lemma 5

PROOF. When updating A , the relevant terms in the loss are

$$\lambda \|\mathcal{X}_{old} - \llbracket A, B, C_{old} \rrbracket\|_F^2 + \|\mathcal{X}_{new} - \llbracket A, B, C_{new} \rrbracket\|_F^2.$$

Taking the derivative with respect to A leads to

$$\begin{aligned} & \lambda(P_{1;old}) + X_{new;(1)}(C_{new} \odot B) \\ & - A[\lambda(Q_{1;old}) + (C_{new}^\top C_{new} * B^\top B)] = 0. \end{aligned}$$

Rewriting the above in matrix form and solving for A yields

$$A \leftarrow P_{1;new}(Q_{1;new})^{-1},$$

where the updated auxiliary matrices $P_{1;new}$ and $Q_{1;new}$ include contributions from both old and new data, as stated in the lemma.

Analogously, the update for B relies on minimizing

$$\lambda \|\mathcal{X}_{old} - \llbracket A, B, C_{old} \rrbracket\|_F^2 + \|\mathcal{X}_{new} - \llbracket A, B, C_{new} \rrbracket\|_F^2,$$

with respect to B . By taking the derivative with respect to B and setting it to zero, we arrive at

$$\begin{aligned} & \lambda(P_{2;old}) + X_{new;(2)}(C_{new} \odot A) \\ & - B[\lambda(Q_{2;old}) + (C_{new}^\top C_{new} * A^\top A)] = 0. \end{aligned}$$

Reorganizing terms leads to

$$B \leftarrow P_{2;new}(Q_{2;new})^{-1}.$$

For D in the matrix-tensor streaming setting, consider the part of the objective that couples C_{new} (as well as C_{old}) to Y_{old} and Y_{new} :

$$\mu \|Y_{old} - C_{old}D^\top\|_F^2 + \|Y_{new} - C_{new}D^\top\|_F^2.$$

Taking the derivative with respect to D and setting it to zero yields

$$-2(\mu C_{old}^\top(Y_{old} - C_{old}D^\top) + C_{new}^\top(Y_{new} - C_{new}D^\top)) = 0.$$

Rewriting this and collecting terms gives

$$\mu C_{old}^\top Y_{old} + C_{new}^\top Y_{new} = (\mu C_{old}^\top C_{old} + C_{new}^\top C_{new})D^\top.$$

Thus,

$$D^\top = (\mu Q_{3;old} + C_{new}^\top C_{new})^{-1} (\mu P_{3;old} + Y_{new}^\top C_{new})^\top,$$

and equivalently, we write

$$D \leftarrow P_{3;new}(Q_{3;new})^{-1},$$

which completes the proof. $\square$

A.6 Proof of Theorem 6

PROOF. The time complexity of FOCAL corresponds to the sum of the computational costs for updating the factor matrices A, B, C , and D . Assume that K_{new} and R are sufficiently small compared to I and J . For updating C , the dominant term is $O(IJK_{new}R + K_{new}LR)$ in Lemma 4. The updates for A and B require $O(IJK_{new}R)$ operations, and for the matrix D , the update has a computational cost of $O(LR^2 + K_{new}LR)$ according to Lemma 5. By summing the costs for all four factor matrices, the overall time complexity is given by $O(IJK_{new}R + (K_{new} + R)LR)$.

Next, storing A, B, C_{new} , and D requires a total of $R(I + J + K_{new} + L)$ space. The tensor X_{new} and the matrix Y_{new} require IJK_{new} and $K_{new}L$ spaces, respectively. We need $O(IJK_{new} + R(I + J + K_{new} + R) + (I + J)K_{new}R + L(R + K_{new}))$ space to compute the auxiliary matrices P_1, P_2, P_3 and Q_1, Q_2, Q_3 . Thus, assuming $K_{new}, R \ll I, J$, we obtain a total of $O(IJK_{new} + K_{new}L + LR)$ space complexity. $\square$

B Additional Experiments

To verify the effect of our frequency filter, we evaluate on synthetic data that deliberately exhibit clear seasonal patterns. We compare FOCAL against SOFIA [29], a competitive method that accepts the seasonal period as side information and leverages it to improve accuracy. Because SOFIA benefits from known periods while our method discovers and adapts to them via a learned frequency filter, this setting is a principled and informative benchmark.

We construct a coupled dataset (X, Y) with shared mode-1 factors. The tensor $X \in \mathbb{R}^{20 \times 20 \times 3000}$ has total time length 3000. We reveal the stream by providing an initial block X_{init} of length 300 and then appending $K_{new} = 30$ slices per update. The coupled matrix $Y \in \mathbb{R}^{20 \times 40}$ does not evolve over time.

All non-temporal factors are drawn i.i.d. from $\mathcal{N}(0, 1)$. The target rank is set to $R = 2$. The temporal factor for each rank-1 component is a sinusoid with a *piecewise* period: for time indices $t = 0, \dots, 2999$,

- **First half:** periods $p \in \{2.5, 5\}$, i.e.,

$$\sin(2\pi t/2.5 + \phi_1), \quad \sin(2\pi t/5 + \phi_2),$$

- **Second half:** periods $p \in \{3, 13\}$, i.e.,

$$\sin(2\pi t/3 + \phi_3), \quad \sin(2\pi t/13 + \phi_4),$$

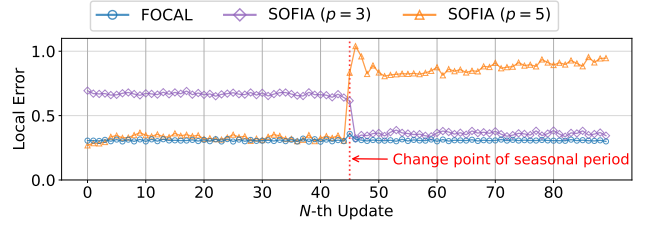


Figure 9: Local reconstruction error per update on a stream with piecewise seasonality (periods $\{2.5, 5\} \rightarrow \{3, 13\}$). FOCAL maintains uniformly low error with only a brief bump at the change point, whereas SOFIA fits only the half whose fixed period ($p = 5$ or $p = 3$) matches the active season.

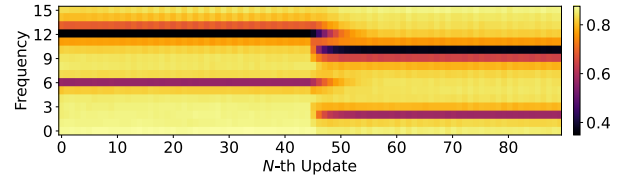


Figure 10: Learned frequency filter over time. FOCAL automatically lowers the penalty around the ground-truth frequencies without any side information, confirming that the filter adapts to evolving seasonality in real time.

where independent random phases ϕ_i are added to each component. Noise is injected into both X and Y : we add zero-mean i.i.d. AWGN with power chosen to achieve SNR = 10 dB. Both methods use target rank $R = 2$ and are initialized with the first 10% of the stream.

- **FOCAL.** The frequency filter ϕ_X is initialized to zero, the forgetting factor is set to 0.9, and the learning rate to 0.02.
- **SOFIA.** We provide the period p as a hyperparameter and evaluate two settings: $p = 3$ (matched to the *second* half of the stream) and $p = 5$ (matched to the *first* half). SOFIA does not include a mechanism to detect or adapt to mid-stream period changes.

As illustrated in Figure 9, FOCAL consistently achieves the lowest error across the entire stream. SOFIA ($p = 3$) fits the second half well but incurs large errors in the first half; SOFIA ($p = 5$) exhibits the opposite behavior—accurate in the first half and sharply degraded thereafter. FOCAL shows a brief increase in error at the change point but rapidly adapts and returns to low error levels, indicating its robustness to evolving seasonal structure.

Figure 10 visualizes the frequency filter ϕ that FOCAL learns over time. Using the mapping $f = K_{new}/p$ (with $K_{new} = 30$), FOCAL reduces the penalty at frequencies aligned with the active periods: 12 and 6 during the first half ($p = 2.5, 5$), and 10 and ≈ 2.3 during the second half ($p = 3, 13$). The filter thus tracks the seasonal components without any prior knowledge of the periods.

These results highlight a key advantage of FOCAL. Unlike SOFIA—which relies on a fixed, correctly specified period and lacks a change-detection mechanism—FOCAL discovers and updates temporal patterns online via its learned frequency filter. As a result, FOCAL maintains low error both before and after shifts in seasonality, demonstrating accurate decomposition *without* side information and *despite* non-stationary temporal dynamics.